# Segmenting Metastatic Prostate Cancer in Whole-Body PSMA PET Scans with a Hybrid CNN-Mamba Network

**James Wigfield**

Supervised by Dr Jake Kendrick and Dr Mubashar Hassan

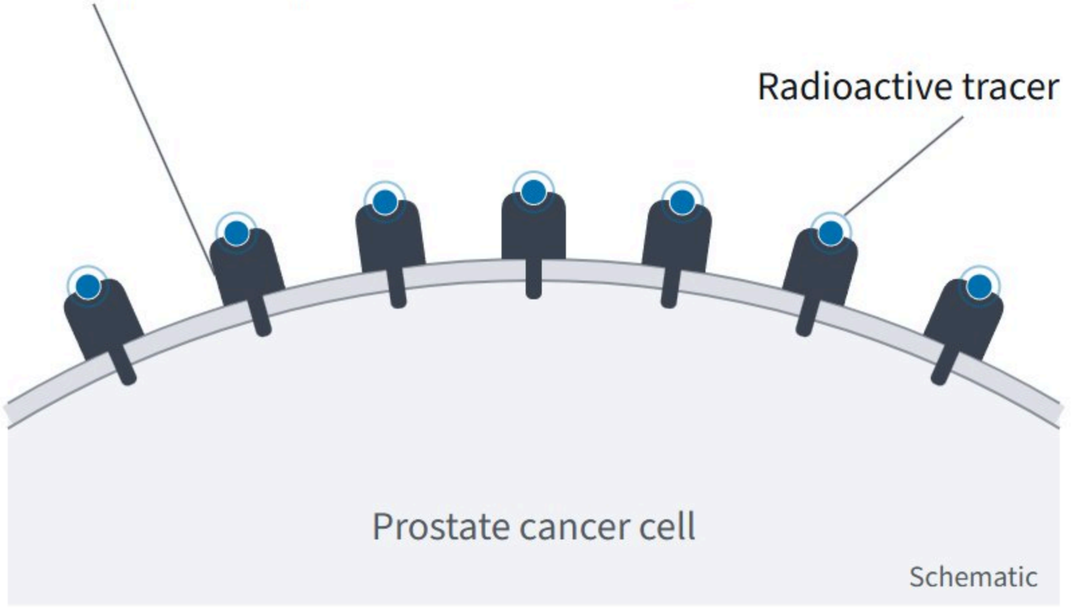


# Metastatic Prostate Cancer and PSMA PET/CT

## 35% or less

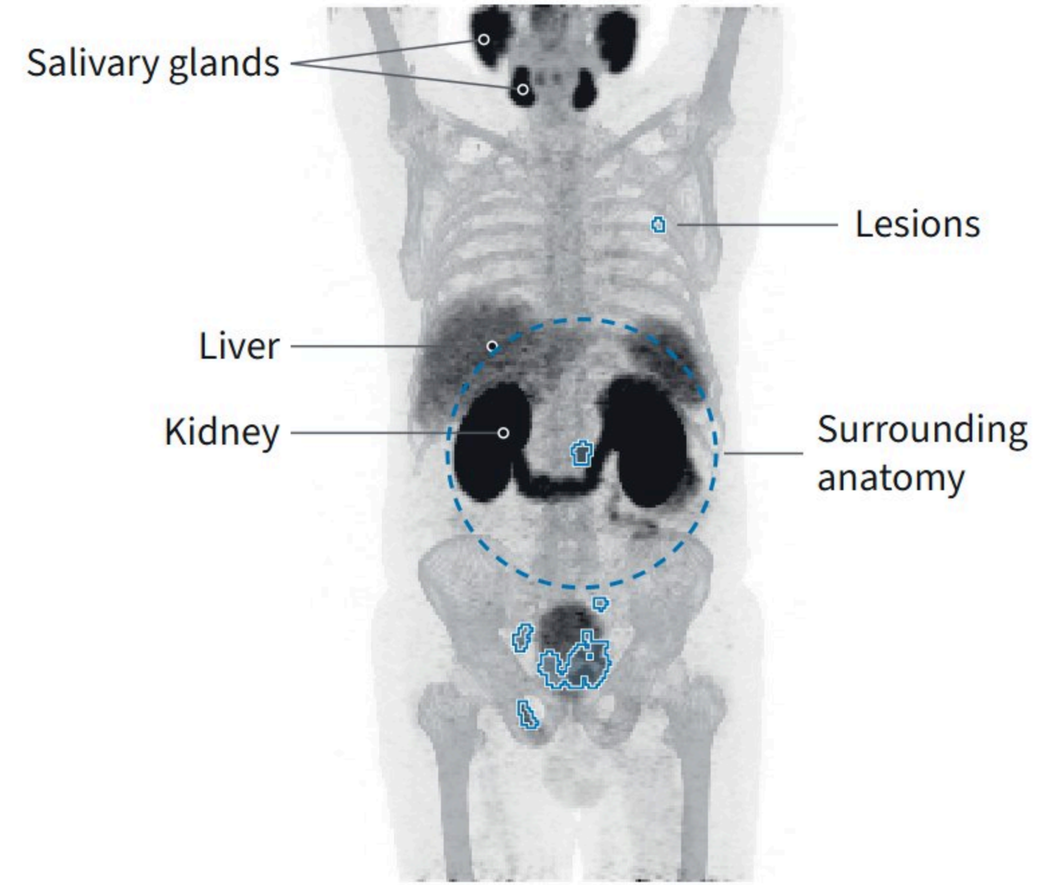
five-year survival once it spreads

**PSMA** (prostate-specific membrane antigen)




NORMAL UPTAKE

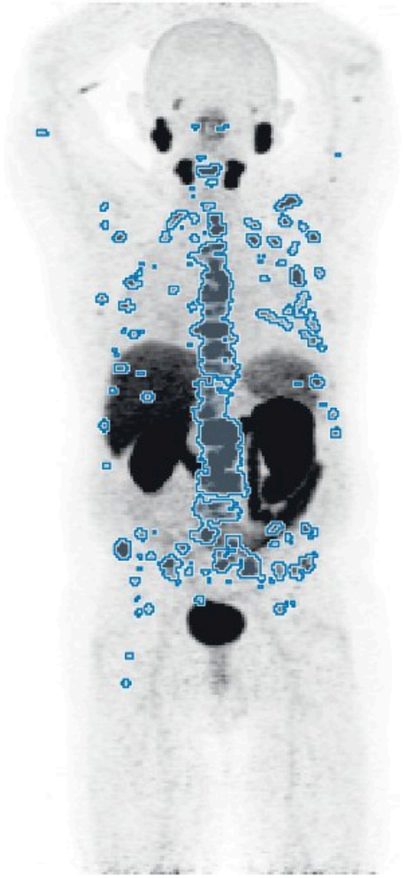
**PSMA PET/CT**




[1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022 · [2] Sheikhbahaei et al., Eur J Nucl Med Mol Imaging 2017

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

# Manual Outlining and Expert Agreement

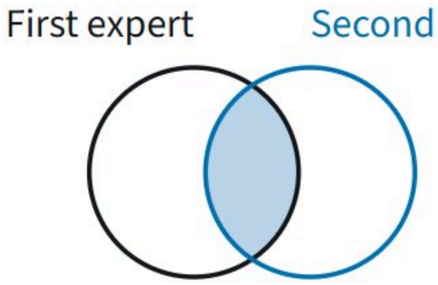


One scan from this project's data

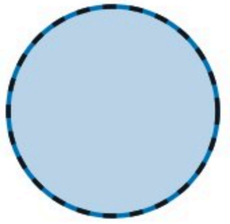
**Over 100 lesions**
in some scans

2022 study, Kendrick et al.

## Two experts, the same 28 scans




**33%**
Dice, on average



**100%**
Identical outlines

Second expert found about 7 out of 10




[1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

# PSMASegmentator and the Patch Size Limit

Open-source tool by Joel Noble, PhD student,
UWA Medical Physics Research Group

Built on nnU-Net, a CNN framework
that configures itself to any new dataset

Trained on 1,035 cases

3D CNNs can't see the entire scan at once

**GPU memory limits patch size**

☐ PSMASegmentator's patch, to scale: 288 x 128 x 128 voxels

Coronal view; front to back not shown




[3] Isensee et al., Nature Methods 2021

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

# Mamba, a Selective State Space Model

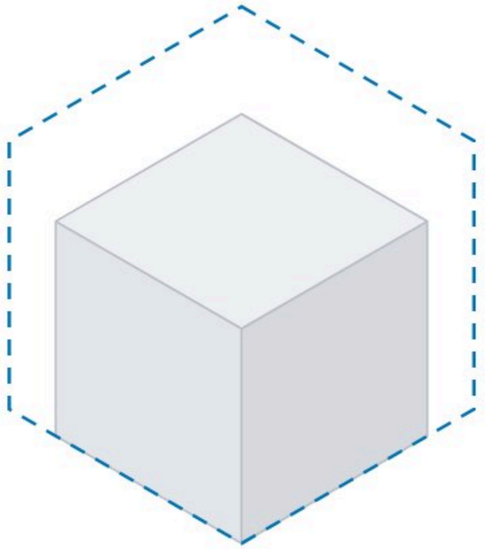
One step at a time, with a fixed-size memory

Selective: kept ignored



Flattened into a long 1D sequence

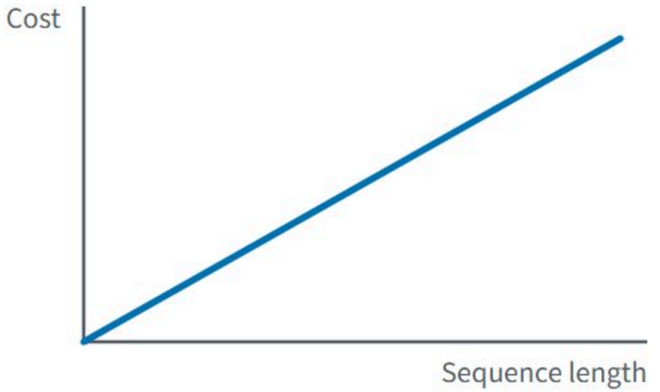
Views across the whole patch



3D image features

Could free memory for a bigger patch

Cost grows linearly with sequence length




Schematic



[4] Gu and Dao, arXiv 2023

# Research Question

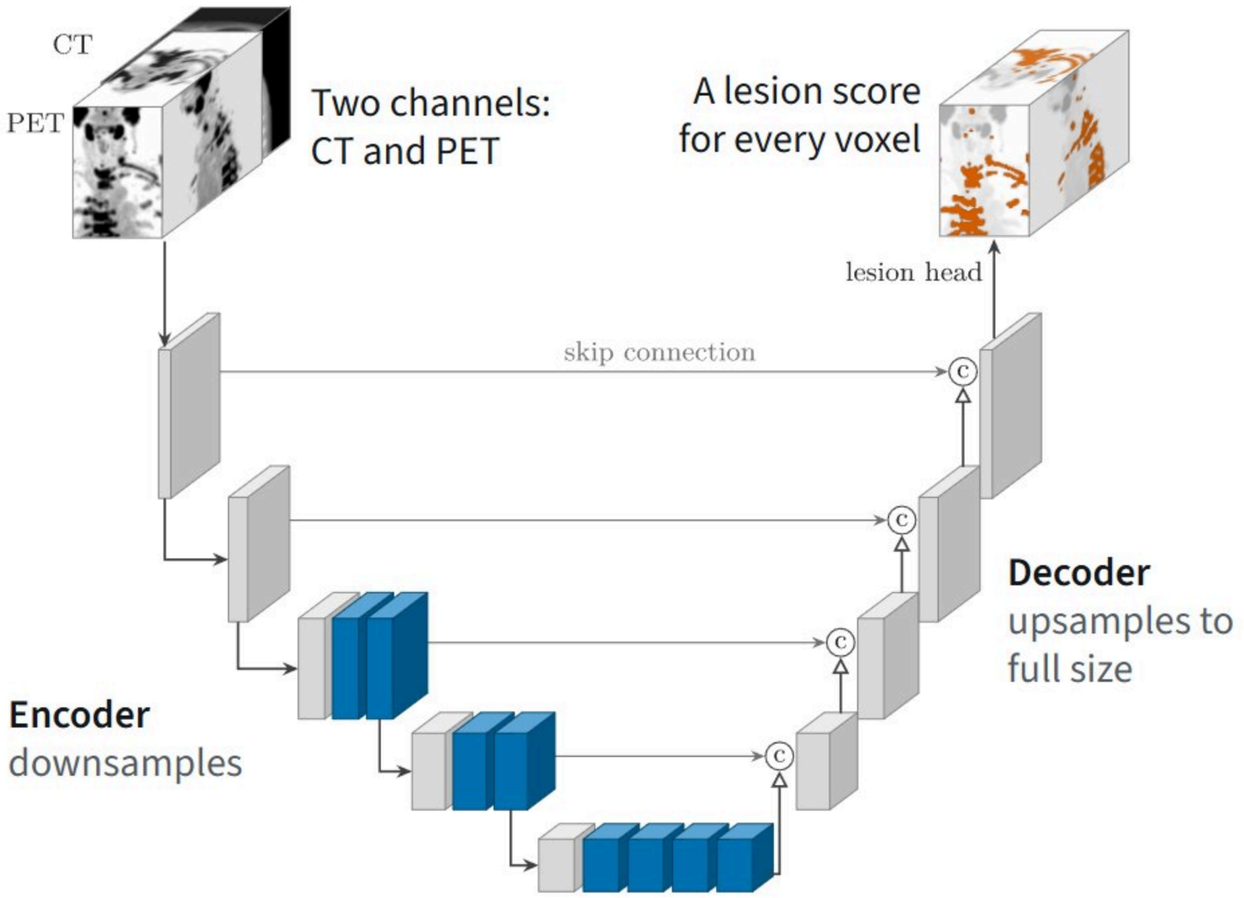
Hybrid CNN-Mamba vs PSMASegmentator

Does a hybrid CNN-Mamba network find lesions more effectively than PSMASegmentator on scans neither was trained on?

Hybrid CNN-Mamba vs CNN-only version

Do the Mamba blocks improve lesion detection, compared with a CNN-only version of the same network?

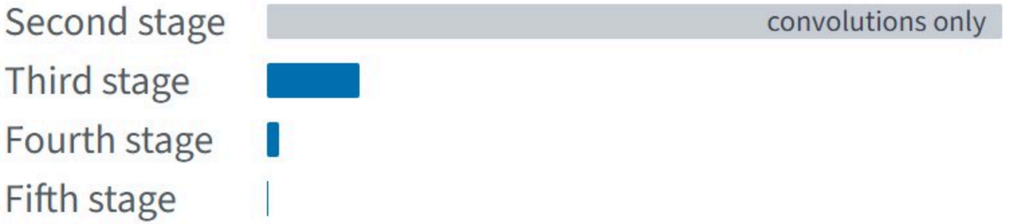
# Hybrid CNN-Mamba Network Architecture




Convolutions in every stage

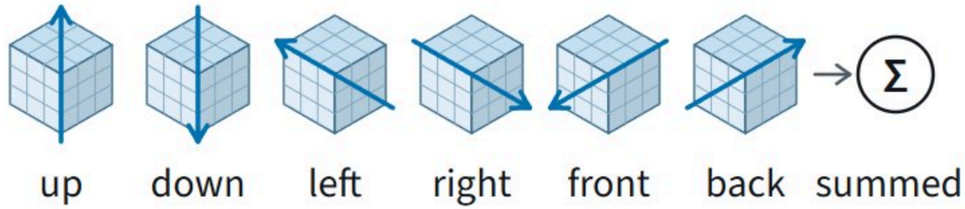
Mamba blocks only in the three deepest stages

**Flattened sequence length, to scale**




About eight times shorter at each stage

**Each Mamba block: six directions, summed**



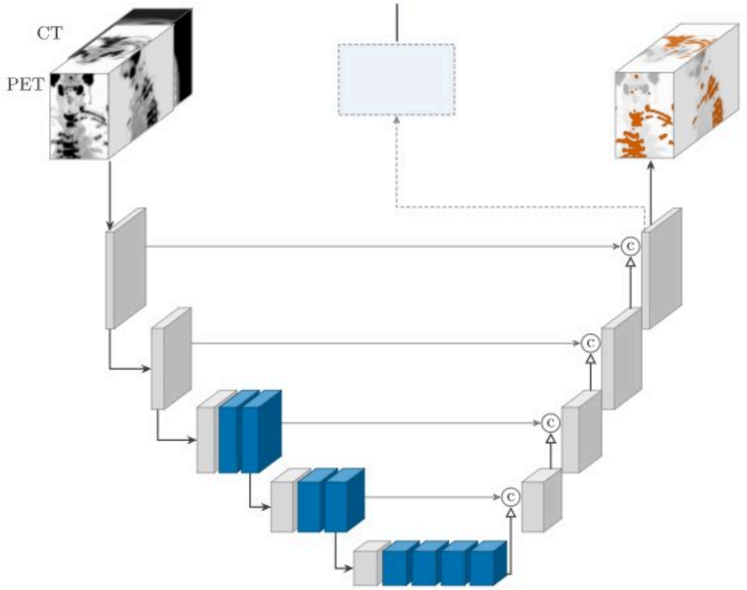

Schematic

[5] Xing et al., IEEE Trans Med Imaging 2026 · [6] Xing et al., MICCAI 2024

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

# Organ Head and Five-Model Ensemble

**Second output**, used during training only
outlines 24 anatomical structures




**Earlier version**

Flagged lesions in all seven scans that had no lesions

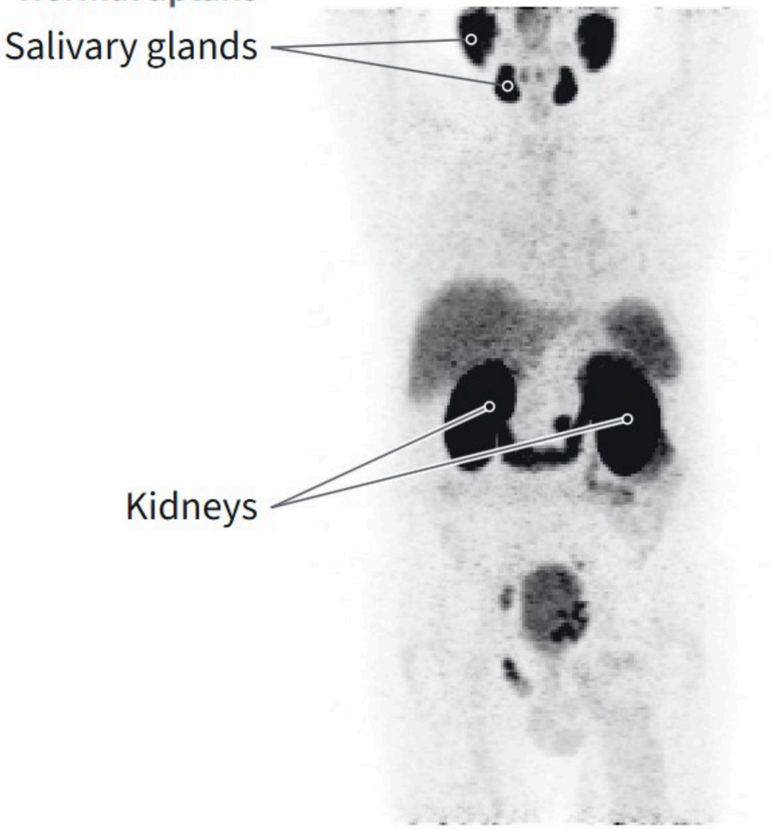
●●●●●●●○○○○○○○○○○○○○○

**21 → 7**

before — with the organ head

false alarms on those scans
with nothing else changed

**Normal uptake**




**Final model: five versions**



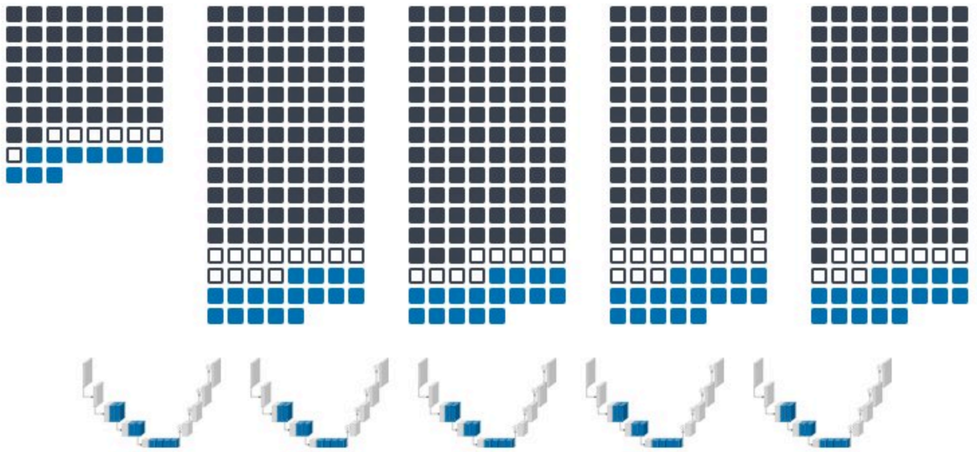
each trained on a different portion of the data
averaged voxel by voxel; threshold 0.5
predicted lesions under 10 voxels removed

[7] Rokuss et al., arXiv 2024 · [8] Wasserthal et al., Radiol Artif Intell 2023 · [2] Sheikhbahaei et al., Eur J Nucl Med Mol Imaging 2017
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

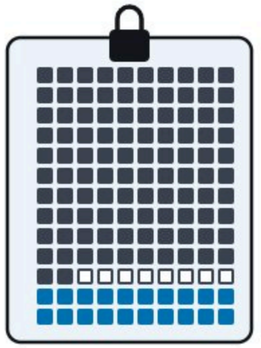
# Datasets, Test Set and Scoring

- **autoPET-III:** 597 scans, ▢ 58 with no lesions
  labels: SUV threshold, then corrected by an expert
- **DEEP-PSMA:** 100 scans, advanced disease, every scan has lesions
  labels: SUV threshold, normal uptake removed by a physician

**The other 567:** five-fold cross-validation



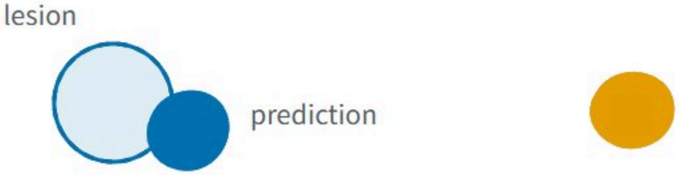
the five versions in the ensemble



**130 sealed away**
never trained on, scored once

## Scoring

**My model** and **PSMASegmentator** used the same scoring script
written by Joel Noble, used with his permission




**Found:** a lesion that any predicted voxel touches

**False alarm:** a prediction touching no lesion

Schematic

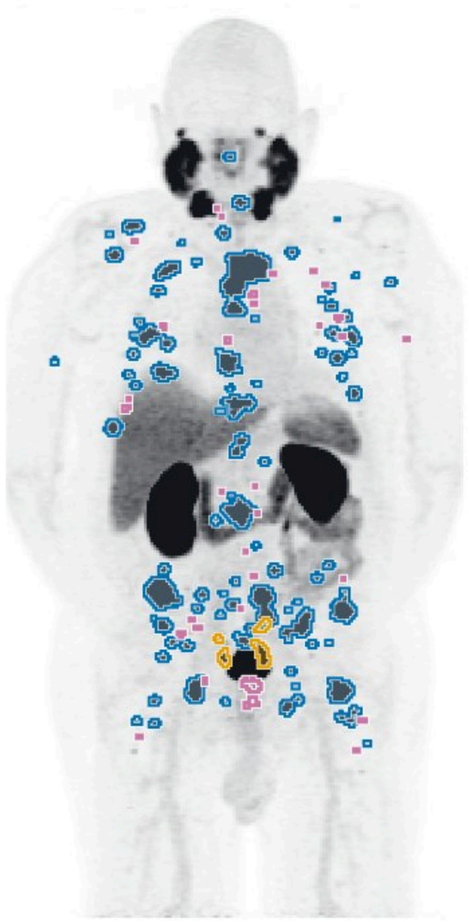
**F1** balances lesions found against false alarms

**Dice** shows how well the outlines match



[9] Jeblick et al., Sci Data 2026 · [10] Jackson et al., DEEP-PSMA design document 2025

# Results on the Sealed Test Set



Sealed test set, 130 scans

**F1 84.3%**

**82%** Found

**86%** Flagged and real

**59.6%** Dice
over the 122 scans with lesions

- Found
- Missed
- False alarm

The scan: one sealed DEEP-PSMA test scan, its cohort's median case by F1

**autoPET-III**, 110 scans

| | |
|---|---|
| F1 | 84.3% |
| Found | 85.9% |
| Flagged and real | 82.7% |

PSMASegmentator trained on these scans

**DEEP-PSMA**, 20 scans

| | |
|---|---|
| F1 | 84.3% |
| Found | 75.1% |
| Flagged and real | 96.0% |

My model trained on 80 of the 100

**Neither dataset here gives a fair comparison**



Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

# Comparison with PSMASegmentator

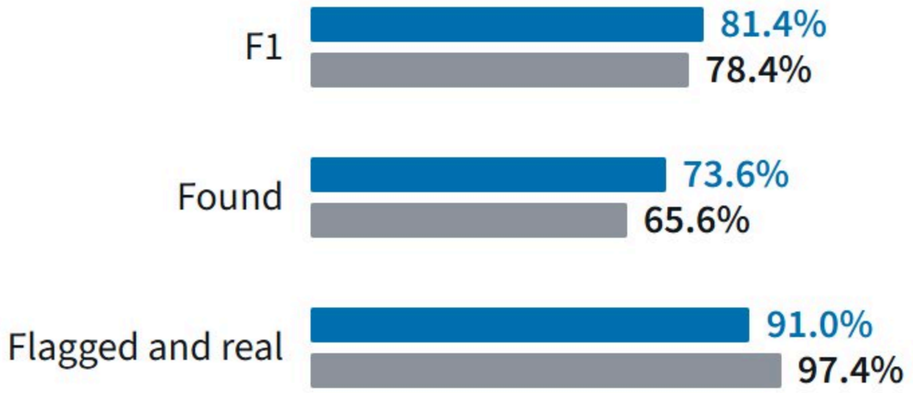
**Hybrid CNN-Mamba**
a single version,
trained only on autoPET

430 training scans
one model

**PSMASegmentator**

1,035 training scans
five models, averaged

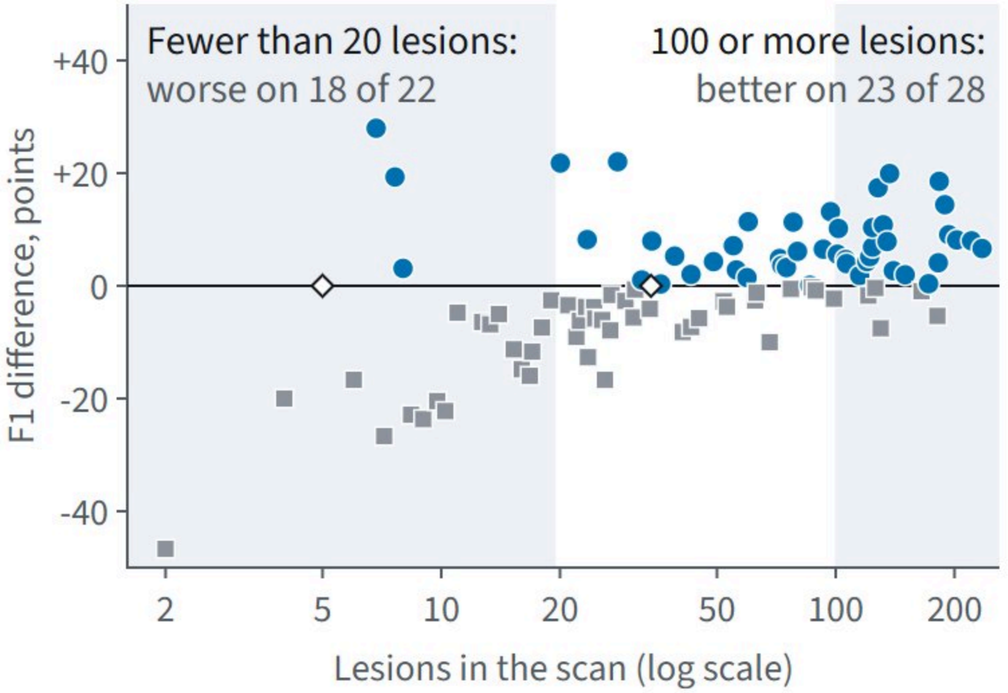
**Pooled over 100 DEEP-PSMA scans neither model trained on**




**+3.0 points**
95% interval 1.2 to 4.7

**Scan by scan, hybrid minus PSMASegmentator**

Better on 48 scans, worse on 50, tied on 2



# Mamba Blocks and Patch Size

● Hybrid CNN-Mamba

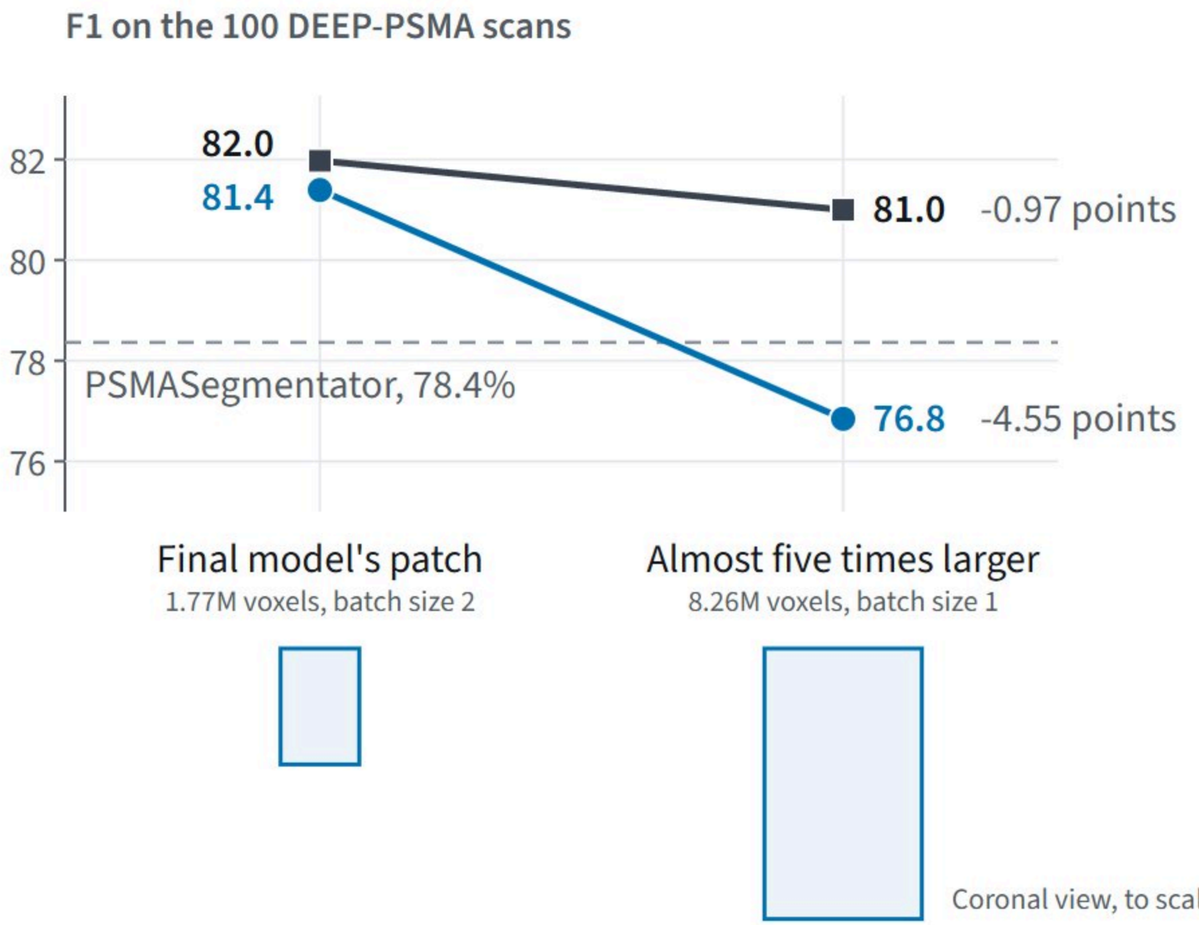
■ CNN-only version

Same network, every Mamba block removed
Both trained only on autoPET

**At the final model's patch**

No clear difference on DEEP-PSMA
autoPET test set (secondary): hybrid +1.79 points

**Gain over PSMASegmentator not credited to Mamba**



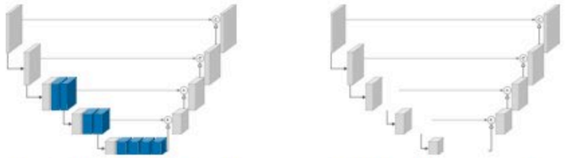

Each model trained once

# Discussion and Limitations

### Gain over PSMASegmentator

**+3.0** points of pooled F1

95% interval 1.2 to 4.7

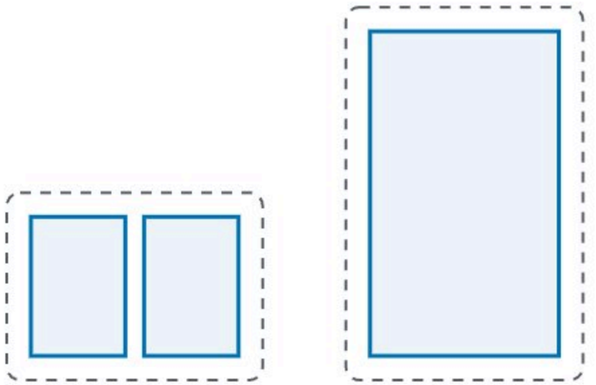


Hybrid CNN-Mamba  CNN-only version

Most likely something both networks share, not Mamba

U-Mamba retest, 2024:
no effect from its Mamba layers against a strong CNN

### Patch size and batch size



**Batch size 2**
final model's patch

**Batch size 1**
almost five times larger

The larger patch still needed batch size 1

### Mean time per scan

one RTX 4090, 20 sealed DEEP-PSMA test scans

Hybrid  PSMASegmentator

**Default settings: 5 against 40 passes**

18.1 s

43.8 s

**Same number of passes**

18.1 s

17.2 s

PSMASegmentator slightly faster

**Peak memory**

5.2 GiB  4.5 GiB

**Limits**

Each model in the CNN-only comparison trained once
against a spread of about 0.6 F1 points between the final model's five versions

Not like-for-like with PSMASegmentator
most setup differences favour PSMASegmentator



[11] Isensee et al., MICCAI 2024

# Conclusion and Next Step

Final model, sealed test set

**F1 84.3%**

**Hybrid CNN-Mamba** vs **PSMASegmentator**

## Scored higher, on scans neither model had seen

single version trained only on autoPET: pooled F1 81.4% against 78.4%
found more lesions, with more false alarms

**Hybrid CNN-Mamba** vs **CNN-only version**

## A CNN-only version did about as well, so the credit doesn't go to Mamba

F1 82.0% for the CNN-only version, 81.4% for the hybrid

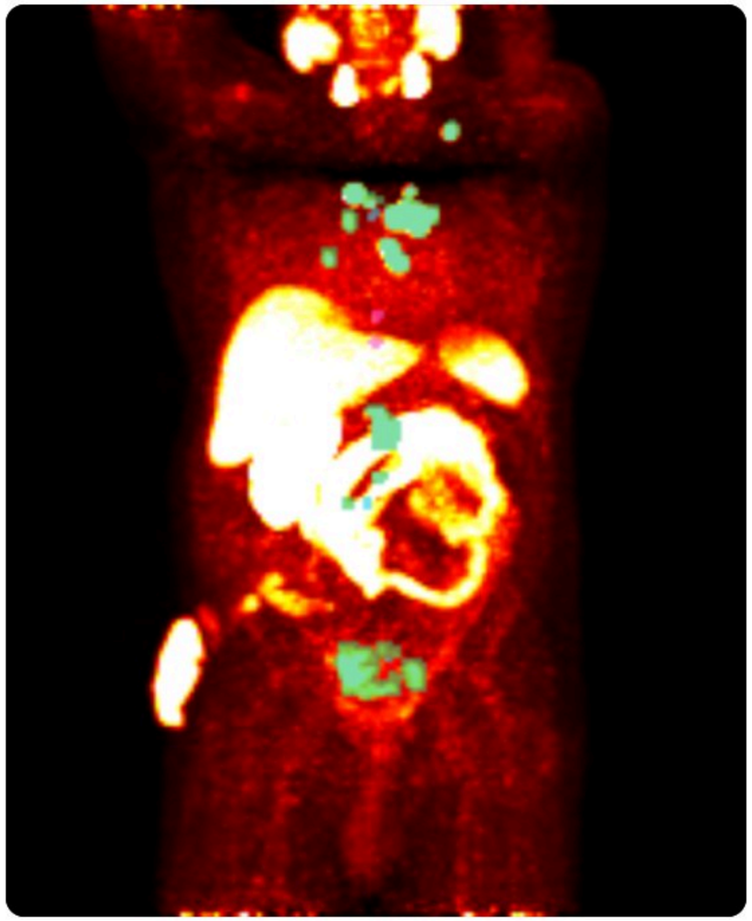
# Thank you

**Next step**

The code will be released open source, so the UWA Medical Physics Research Group can retrain and test it on Sir Charles Gairdner Hospital data

DEMO

# Recorded Run of the Final Model in the App



## The five-model ensemble in my app, on a sealed test scan

Sped up 6x while it runs; the result then turns in the app's 3D view

### Result in the app's 3D view

- Found
- Missed
- False alarm

The app's own colours. Before the run, cyan shows the ground truth.

Threshold and size filter as written down beforehand; no test-time augmentation.



Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

# Setup Differences with PSMASegmentator

| Factor | Hybrid, single version | PSMASegmentator | Favours |
|---|---|---|---|
| Training scans | **430 autoPET-III** | 1,035 (597 autoPET-III, 438 Western Australian) | PSMASegmentator |
| Models at inference | **one** | five, averaged | PSMASegmentator |
| Test-time augmentation | **off** | on, eight mirrored passes | PSMASegmentator |
| Deep supervision | **none** | yes | PSMASegmentator |
| PET/CT misalignment augmentation | **none** | yes | PSMASegmentator |
| Training patch | **1.77M voxels** | 4.72M voxels | PSMASegmentator |
| Parameters | **39.7M** | about 102.4M | PSMASegmentator |
| Optimiser | **AdamW, cosine decay** | SGD, polynomial decay | unclear |
| Post-processing | **none** | disabled for this comparison | neither |
| Matching rule and scoring grid | **same script and grid** | same script and grid | neither |

**Most factors favour PSMASegmentator, so the gain is more likely to understate than overstate the recipe; none was tested**

# Paired Differences on the 100 DEEP-PSMA Scans

| Hybrid minus PSMASegmentator | Points | 95% interval |
|---|---|---|
| Pooled F1 | **+3.0** | 1.2 to 4.7 |
| Pooled found (sensitivity) | **+8.1** | 5.9 to 10.2 |
| Pooled flagged and real (PPV) | **-6.4** | -7.9 to -5.2 |
| Per-scan F1, mean | **-0.6** | -2.8 to 1.6 |
| Per-scan Dice, mean | **+6.4** | 4.3 to 8.5 |

Per scan: better on 48, worse on 50, tied on 2

Fewer than 20 lesions: worse on 18 of 22; 100 or more: better on 23 of 28

Paired bootstrap over patients; pooled F1 weighs lesion-heavy scans the most

# False Alarms on Scans with No Lesions

**5** of the 8 sealed test scans with no lesions were flagged
by the final five-model ensemble

**6 to 8** each version alone, before averaging
the ensemble flagged fewer than any version alone

**21 to 7** false alarms on the seven validation scans with no lesions
an earlier version, before and after the organ head

**First change to try: the organ head at every deep-supervision level, as PSMASegmentator has**

Here the organ head is single-scale, with no post-processing behind it

# Lesion Matching Rule and Dice

### The rule

**Found:** a lesion that any predicted voxel touches

**False alarm:** a predicted component touching no lesion

The same scoring script for both models, on the same grid

written by Joel Noble, used with his permission

The rule is blind to over-segmentation, so Dice is reported beside every lesion figure

### Dice on the 100 DEEP-PSMA scans

**62.3%** Hybrid, single version

**55.9%** PSMASegmentator

+6.4 points (4.3 to 8.5), consistent with finding more lesions without smearing

# Labels of the Two Datasets

**autoPET-III**

SUV threshold within drawn volumes of interest, then corrected slice by slice by an expert

Sealed test, 110 scans: found 85.9%, flagged and real 82.7%

**DEEP-PSMA**

SUV of at least 3, physiological uptake removed by a nuclear medicine physician

About 68 separate pieces per scan

Sealed test, 20 scans: found 75.1%, flagged and real 96.0%

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**Possible explanation, not tested**

A model trained mostly on expert-corrected labels covers the bulk of each lesion but misses many small thresholded pieces

Nearly everything it predicts then lands on thresholded disease

The final model trained on 80 of the 100 DEEP-PSMA scans; Dice is not comparable across the two

# Mamba Blocks in the Three Deepest Stages

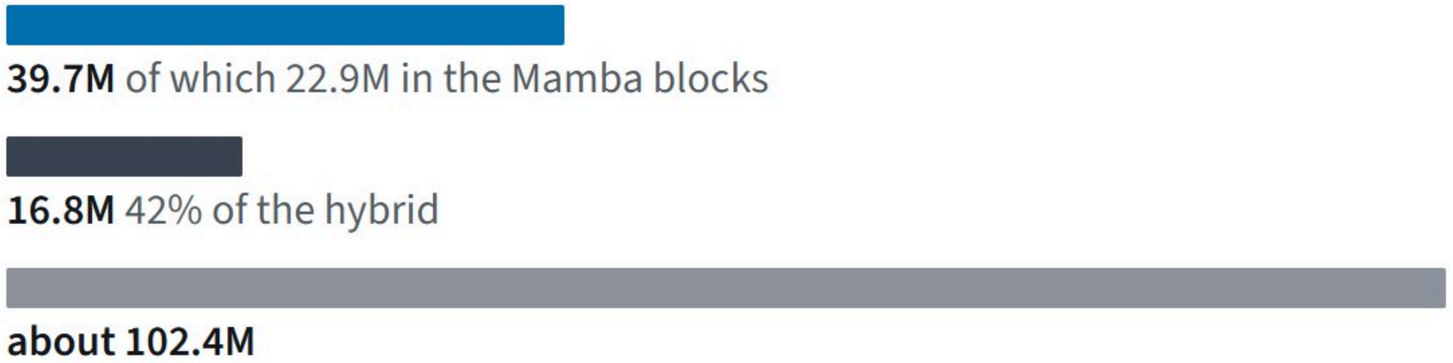
| Stage | Channels | Sequence length per direction | Blocks |
|---|---|---|---|
| 1 | 32 | 1,769,472 | convolutions |
| 2 | 64 | 221,184 | convolutions |
| 3 | 128 | **27,648** | **convolutions and 2 Mamba blocks** |
| 4 | 256 | **3,456** | **convolutions and 2 Mamba blocks** |
| 5 | 320 | **432** | **convolutions and 4 Mamba blocks** |

The shallow stages hold low-level texture, which convolution handles more cheaply than a scan of that length

Each Mamba block scans six directions and sums them; neither choice was tested against an alternative

# Size of the CNN-Only Version

**Parameters**




Capacity is a confound in the comparison with PSMASegmentator

In the CNN-only comparison the parameter gap is the same at both patch sizes, so it cannot on its own make the gap between them change with patch size, if the effects add

# Comparisons in the CNN-Only Experiment, with Intervals

| F1 on the 100 DEEP-PSMA scans | Points | 95% interval | Under the rule |
|---|---|---|---|
| Hybrid minus CNN-only, larger patch | **-4.15** | -5.07 to -3.29 | real |
| Hybrid minus CNN-only, final model's patch | **-0.58** | -1.31 to 0.19 | no clear effect |
| **Difference between those two gaps** | **-3.57** | **-4.72 to -2.48** | **real** |
| Larger patch, hybrid | **-4.55** | -5.68 to -3.45 | real |
| Larger patch, CNN-only version | **-0.97** | -1.78 to -0.08 | no clear effect |

Rule, written down before scoring: the interval excludes zero and the size is at least 1.0 point

One training run per model; the final model's five versions spread by about 0.6 points

Larger patch at batch size 1, the final model's patch at 2; several seeds per model, with the larger patch at batch size 2, would settle both

# Inference Time in the Dissertation and the Abstract

| Mean time per scan, one RTX 4090, 20 sealed DEEP-PSMA test scans | Hybrid | PSMASegmentator |
|---|---|---|
| Default settings: 5 against 40 passes | **18.1 s** | 43.8 s |
| Same passes, 5 each | **18.1 s** | 17.2 s |
| Same passes, 40 each | **131.2 s** | 43.8 s |
| Median, 5 passes each | **11.6 s** | 17.0 s |
| Peak memory | **5.2 GiB** | 4.5 GiB |

| The abstract's home-PC benchmark, one RTX 5070 Ti (not in the dissertation) | Hybrid | PSMASegmentator |
|---|---|---|
| Published settings, mean seconds per scan | **32** | 63 |
| Same passes, 5 each, mean seconds | **32** | 33 |

**The default gap comes from PSMASegmentator's test-time augmentation, not the architecture**

# Cross-Validation and the Five Folds

**567** scans in the cross-validation pool, in five folds split by patient
67 in the first fold, 125 in each of the others

**81.8%** pooled F1 over the 567 scans, each scored by the version that never trained on it
fold-to-fold standard deviation 1.5 points

**84.3%** F1 of the five versions averaged, on the sealed test set
one version alone: 83.1 on average (standard deviation 0.6)

# Test-Set Results by Dataset with Intervals

| Sealed test set | F1 | Found | Flagged and real |
|---|---|---|---|
| **Pooled, 130 scans** | **84.3%**<br>81.4 to 86.5 | **82.4%**<br>77.4 to 86.5 | **86.2%**<br>83.4 to 88.9 |
| autoPET-III, 110 scans | **84.3%**<br>80.2 to 86.8 | 85.9%<br>79.3 to 90.3 | 82.7%<br>78.8 to 85.9 |
| DEEP-PSMA, 20 scans (in-domain) | **84.3%**<br>79.8 to 87.9 | 75.1%<br>69.2 to 80.3 | 96.0%<br>93.3 to 98.0 |

95% intervals from a bootstrap over patients, added after the test set was scored; no point estimate changed

In-domain: the final model trained on 80 of the 100 DEEP-PSMA scans

# References

[1] J. Kendrick, R. J. Francis, G. M. Hassan, P. Rowshanfarzad, J. S. L. Ong, and M. A. Ebert, "Fully automatic prognostic biomarker extraction from metastatic prostate lesion segmentations in whole-body [68Ga]Ga-PSMA-11 PET/CT images," European Journal of Nuclear Medicine and Molecular Imaging, vol. 50, pp. 67-79, 2022.

[2] S. Sheikhbahaei, A. Afshar-Oromieh, M. Eiber et al., "Pearls and pitfalls in clinical interpretation of prostate-specific membrane antigen (PSMA)-targeted PET imaging," European Journal of Nuclear Medicine and Molecular Imaging, vol. 44, pp. 2117-2136, 2017.

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